

PKU–Zurich PhD Summer School

on Machine Learning for Macroeconomics and Finance

Reading List

Beijing, China · July 6–10, 2026

Admitted participants come from economics, computer science, mathematics, and related fields. To make the most of the week, please arrive with a shared foundation: work through the **pre-reading and pre-coding** below before the school, and use the per-day lists to prepare for, and follow up on, each set of lectures. For each entry, *core* readings are the papers the lectures are built on; *background* items are foundational references that help if the topic is new to you.

Before the Summer School (Pre-Reading & Pre-Coding)

Everyone should complete these two items before arriving in Beijing.

- **Reading — heterogeneous-agent models.** Dirk Krueger, *Heterogeneous Agent Macroeconomics* (lecture notes), 2025. [\[link\]](#)
Read **Chapter 6**, “The SIM in General Equilibrium,” carefully; use **Chapter 4**, “The SIM in Partial Equilibrium,” as a background reference.
- **Coding — neural networks for economic models.** Simon Scheidegger, *Deep Equilibrium Nets: Brock–Mirman tutorials* (Jupyter notebooks).
Read and run both notebooks: Notebook 1 — [deterministic Brock–Mirman](#), and Notebook 2 — [stochastic Brock–Mirman](#). They introduce the deep-equilibrium-net idea on a very simple model with the neural-network scaffolding written out end to end.

Day 1 — Deep Learning Basics

Serguei Maliar (Santa Clara University)

- Maliar, Maliar & Winant (2021). *Deep learning for solving dynamic economic models*. Journal of Monetary Economics 122, 76–101. [\[link\]](#)
The “Maliar–Maliar–Winant method” named in the lectures: it recasts lifetime-reward, Bellman, and Euler-equation problems as deep-learning objectives trained by stochastic gradient descent, using the all-in-one expectation/integration operator. The core method paper for Day 1.
- Maliar, Maliar & Winant (2019). *Will Artificial Intelligence Replace Computational Economists Any Time Soon?*. CEPR Discussion Paper DP14024. [\[link\]](#)
The longer working-paper treatment, with fuller exposition of the all-in-one integration trick, neural-network model reduction, and implementation details useful for the hands-on tutorial.

Recommended background

- Goodfellow, Bengio & Courville (2016). *Deep Learning*. MIT Press (full text free online). [\[link\]](#)
The canonical deep-learning textbook. Skim Part I and Chapters 6–8 (feedforward networks, regularization, optimization/SGD).

- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving*. Quarterly Journal of Economics 109(3), 659–684. [\[link\]](#)
The foundational discrete-time, incomplete-markets heterogeneous-agent model behind the discrete-time HA lecture.
- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy*. Journal of Political Economy 106(5), 867–896. [\[link\]](#)
The classic heterogeneous-agent model with aggregate risk; its curse of dimensionality is exactly what the deep-learning solvers are designed to overcome.

Day 2 — Deep Equilibrium Nets

Simon Scheidegger (HEC Lausanne) and Felix Kübler (University of Zurich)

- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets*. International Economic Review 63(4), 1471–1525. [\[link\]](#)
The flagship paper the DEQN lectures are built on: neural networks trained unsupervised to satisfy equilibrium conditions along simulated paths, with heterogeneity, uncertainty, and occasionally binding constraints. Reproduction code at github.com/sischei/DeepEquilibriumNets.
- Scheidegger & Bilionis (2019). *Machine learning for high-dimensional dynamic stochastic economies*. Journal of Computational Science 33, 68–82. [\[link\]](#)
Combines Gaussian-process regression with active subspaces to solve models up to roughly 500 dimensions; bridges the deep-learning and Gaussian-process halves of Day 2.
- Rasmussen & Williams (2006). *Gaussian Processes for Machine Learning*. MIT Press (full text free online). [\[link\]](#)
The canonical reference for Kübler’s Gaussian-processes and Bayesian numerical-methods lecture; focus on Chapters 2–5 (GP regression, covariance functions, hyperparameter learning).

Day 3 — DeepHAM and Structural Reinforcement Learning

Yucheng Yang (University of Zurich), Benjamin Moll (LSE), Simon Scheidegger / Felix Kübler

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks*. Quantitative Economics 17(2), 297–341. [\[link\]](#)
The DeepHAM method: neural networks approximate value and policy functions and represent the cross-sectional distribution via learned generalized moments. Focus on the moment representation and the simulation-based training objective.
- Yang, Wang, Schaab & Moll (2025). *Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics*. arXiv:2512.18892. [\[link\]](#)
The basis for Moll’s special online lecture: a structural RL method that uses low-dimensional prices as state variables and lets agents learn equilibrium price dynamics from simulated paths (Krusell–Smith, Huggett with aggregate shocks, HANK).
- Chen, Didisheim & Scheidegger (2026). *Deep Surrogates for Finance: With an Application to Option Pricing*. Journal of Financial Economics 177, 104222. [\[link\]](#)

Underpins the “deep surrogate models and uncertainty quantification” lecture: neural-network surrogates approximate expensive structural models to accelerate estimation by orders of magnitude.

Day 4 — Deep Learning and Continuous-Time Macro-Finance

Goutham Gopalakrishna (*University of Toronto, Rotman*)

- Gopalakrishna & Wu (2021). *ALIENs for Continuous Time Economies*. Swiss Finance Institute Research Paper No. 21-34. [\[link\]](#)
The instructor’s flagship method: Active-Learning-Inspired Equilibrium Nets solve high-dimensional, nonlinear continuous-time macro-finance models by combining time-stepping with active learning over the economically important regions of the state space.
- Wu, Guo, Gopalakrishna & Poulos (2024). *Deep-MacroFin: Informed Equilibrium Neural Network for Continuous Time Economic Models*. arXiv:2408.10368. [\[link\]](#)
The companion library and method paper; the natural reference for the hands-on tutorial, showing how to encode HJB and equilibrium conditions as neural-network losses. Code at github.com/rotmanfinhub/deep-macrofin.
- Sirignano & Spiliopoulos (2018). *DGM: A deep learning algorithm for solving partial differential equations*. Journal of Computational Physics 375, 1339–1364. [\[link\]](#)
The Deep Galerkin Method behind the “deep learning for PDEs” lecture: a meshfree algorithm that trains a neural network to satisfy a high-dimensional PDE on randomly sampled points.

Recommended background

- Brunnermeier & Sannikov (2014). *A Macroeconomic Model with a Financial Sector*. American Economic Review 104(2), 379–421. [\[link\]](#)
The foundational continuous-time macro-finance model with a financial sector; its full nonlinear crisis dynamics are the canonical problem class these solvers target.
- Achdou, Han, Lasry, Lions & Moll (2022). *Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach*. Review of Economic Studies 89(1), 45–86. [\[link\]](#)
Recasts the Aiyagari–Bewley–Huggett model in continuous time as coupled HJB and Kolmogorov-forward PDEs — the continuous-time HA toolkit underlying the lectures.

Day 5 — Deep Learning for Macro-Finance and Beyond

Jonathan Payne (*Princeton University*); Keynote: Weinan E (*Peking University*)

- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models*. arXiv:2406.13726 (forthcoming, JPE: Macroeconomics). [\[link\]](#)
Introduces Economic-Model-Informed Neural Networks (EMINNs) to compute global solutions of master equations, with the continuous-time Krusell–Smith model as the lead application.
- Payne, Rebei & Yang (2025). *Deep Learning for Search and Matching Models*. Swiss Finance Institute Research Paper No. 25-05. [\[link\]](#)
Underpins the search-and-matching lecture: extends the deep-learning-for-PDE approach to labor-market models with aggregate shocks and heterogeneous agents.

- Gopalakrishna, Gu & Payne (2026). *Institutional Asset Pricing with Segmentation and Household Heterogeneity*. SSRN Working Paper. [\[link\]](#)

The basis for the asset-pricing-with-household-heterogeneity lecture: a segmented heterogeneous-agent macro-finance model solved globally by deep learning, calibrated to Flow of Funds data.

Recommended background

- Han, Jentzen & E (2018). *Solving high-dimensional partial differential equations using deep learning*. Proceedings of the National Academy of Sciences 115(34), 8505–8510. [\[link\]](#)

The deep-BSDE method: the algorithmic ancestor of these economic PDE solvers and a touchstone of Weinan E’s research, bridging both Day-5 speakers.

- Weinan E (2021). *The Dawning of a New Era in Applied Mathematics*. Notices of the American Mathematical Society 68(4), 565–571. [\[link\]](#)

The keynote’s manifesto: an accessible “AI for Science” perspective on machine learning as a new paradigm for applied mathematics and scientific discovery.